

Q1. Find the null to null bandwidth of the following signal.

$$X(f) = \left[\frac{\sin 2\pi f T}{2\pi f T} \right]^2$$

$$\frac{\sin 2\pi f T}{2\pi f T} = 0$$

$$\sin 2\pi f T = 0$$

$$2\pi f T = \sin^{-1}(0)$$

$$2\pi f T = k\pi \quad (k = \pm 1, \pm 2, \dots)$$

$$f = \frac{k}{2T}$$

$$\text{at } k=0, f=0$$

closest is $k=1$ and $k=-1$

$$k=1 \rightarrow f_2 = \frac{1}{2T}$$

$$k=-1 \rightarrow f_1 = -\frac{1}{2T}$$

} First nulls.

$$BW = f_2 - f_1$$

$$\frac{1}{2T} - \left(-\frac{1}{2T}\right) = \boxed{\frac{1}{T}} \rightarrow \text{null to null bandwidth.}$$

Q2. Find maximum null to null bandwidth of the signal:

$$f(x) = \begin{cases} A & \text{if } -T \leq x \leq T \\ 0 & \text{else} \end{cases}$$

such that its energy is constrained to be E_x .

$$E_x = \int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-T}^T A^2 dx$$

$$E_x = A^2 x \Big|_{-T}^T \quad E_x = A^2(2T) \quad \Rightarrow \quad A = \sqrt{\frac{E_x}{2T}}$$

Fourier Transform:

$$F(f) = \int_{-T}^T A e^{-j2\pi f x} dx$$

$$= \frac{A}{-j2\pi f} \left[e^{-j2\pi f T} \right]_T^{-T}$$

$$= \frac{A}{-j2\pi f} \left[e^{-j2\pi f T} - e^{j2\pi f T} \right] \quad \text{Use } e^{-j\theta} - e^{j\theta} = -2j \sin \theta$$

$$= \frac{A}{-j2\pi f} \left[-2j \sin(2\pi f T) \right] \rightarrow \frac{A \sin(2\pi f T)}{\pi f}$$

Finding nulls:

$$F(f) = \frac{A \sin(2\pi f T)}{\pi f} = 0$$

$$2\pi f T = k\pi \quad (k = \pm 1, \pm 2, \pm 3 \dots)$$

$$f = \frac{k}{2T}$$

First nulls
 $k=1, k=-1$

$$f_1 = \frac{1}{2T}, \quad f_2 = -\frac{1}{2T}$$

$$BW = f_1 - f_2$$

$$= \frac{1}{T} \rightarrow \text{null to null bandwidth.}$$

Q3 Find the rectangular equivalent bandwidth of the signal $x(t) = (1/T)\Lambda(t/T)$; $T \geq 0$ and Λ is defined as

$$\Lambda(t) = \begin{cases} 1+t & \text{if } -1 \leq t \leq 0 \\ 1-t & \text{if } 0 \leq t \leq 1 \end{cases}$$

$\Lambda(t)$ is a convolution of two functions.

$$\Lambda(t) = \Pi(t) * \Pi(t)$$

convolution property $\Pi(t) \leftrightarrow \text{sinc}(f)$ leads to

$$\Lambda(t) \leftrightarrow \text{sinc}^2(f)$$

using time scaling property

$$\Lambda(t/T) \longleftrightarrow T \text{sinc}^2(fT)$$

our signal is $\frac{1}{T} \Lambda(t/T)$

$$\frac{1}{T} \Lambda(t/T) \longleftrightarrow \frac{1}{T} \cdot T \text{sinc}^2(fT)$$

$$X(f) = \text{sinc}^2(fT)$$

$$W_{re} = \frac{1}{2 \max_f |X(f)|^2} \int_{-\infty}^{\infty} |X(f)|^2 df$$

$$|X(f)|^2 = \text{sinc}^4(fT)$$

Computing $\max_f |X(f)|^2$

$$\text{sinc}(0) = 1$$

so at $f = 0$

$$|X(0)|^2 = 1^4 = 1$$

$$\max_f |X(f)|^2 = 1$$

$$\text{Computing } E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Finding for 1 side and multiplying by 2 since symmetric triangle.

for $0 \leq t \leq T$

$$-1 \leq \frac{t}{T} \leq 1$$

$-T \leq t \leq T \rightarrow$ triangle got scaled, limits changed.

$$E_x = \int_{-T}^T \left| \frac{1}{T} \Lambda\left(\frac{t}{T}\right) \right|^2 dt \Rightarrow \frac{1}{T^2} \int_{-T}^T \Lambda^2\left(\frac{t}{T}\right) dt$$

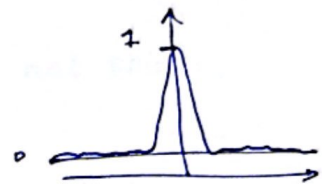
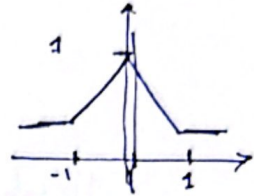
for $0 \leq t \leq T \rightarrow 1 - t/T$

$0 \leq t \leq T \rightarrow$ taking $x = \frac{t}{T}$ $t = Tx$ $dt = Tdx$

$$\text{So } \frac{2}{T^2} \int_0^T \left(1 - \frac{t}{T}\right)^2 dt$$

$$\frac{2}{T^2} \int_0^1 (1-x)^2 T dx$$

non zero from -1 to 1 only.



$$E_x = \frac{2}{T} \int_0^1 (1-x)^2 dx$$

$$\text{at } t=T, x=1$$

$$\int_0^1 (1-2x+x^2) dx$$

$$\frac{2}{T} \left[x - x^2 + \frac{x^3}{3} \right]_0^1$$

$$E_x = \frac{2}{T} \cdot \frac{1}{3} = \frac{2}{3T}$$

$$W_{re} = \frac{\frac{2}{3T}}{2 \max f_{max}} = \frac{1}{3T} \rightarrow \text{rectangular bandwidth.}$$

2. Prove that the Fourier transform of a signal is not unique.
Proceed by first showing

$$\sum_n \delta(t - nT_s) \xleftrightarrow{F} \sum_n e^{-j2\pi f n T_s}$$

and then showing there is another function that is the

$$\text{FT of } \sum_n \delta(t - nT_s)$$

starting from FT definition

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$

$$X(f) = \int_{-\infty}^{\infty} \left(\sum_{n=-\infty}^{\infty} \delta(t - nT_s) \right) e^{-j2\pi f t} dt$$

$$X(f) = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(t - nT_s) e^{-j2\pi f t} dt$$

Using sifting property of delta:

$$\int_{-\infty}^{\infty} \delta(t - a) g(t) dt = g(a)$$

$$\text{Here } a = nT_s, g(t) = e^{-j2\pi f t}$$

$$X(f) = \sum_{n=-\infty}^{\infty} e^{-j2\pi f n T_s}$$

$$\sum_n \delta(t - nT_s) \xleftrightarrow{F} \sum_n e^{-j2\pi f n T_s} \quad \text{proved}$$

(b) A single delta in frequency corresponds to a complex exponential in time:

$$\delta(f - f_0) \xleftrightarrow{F^{-1}} e^{j2\pi f_0 t}$$

$$Y(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta\left(f - \frac{k}{T_s}\right)$$

Taking inverse FT:

$$y(t) = \int_{-\infty}^{\infty} Y(f) e^{j2\pi f t} df$$

$$y(t) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} \delta\left(f - \frac{k}{T_s}\right) e^{j2\pi f t} df$$

$$y(t) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} e^{j2\pi (k/T_s) t}$$

$$\frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta\left(f - \frac{k}{T_s}\right) \xleftrightarrow{F^{-1}} \frac{1}{T_s} \sum_{k=-\infty}^{\infty} e^{j2\pi k t / T_s}$$

$$\sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} e^{j2\pi k t / T_s}$$

impulse train in time equals the exponential

series, so inverse FT of $Y(f)$ is exactly $x(t)$

so:

$$\sum_n \delta(t - nT_s) \xleftrightarrow{F} \frac{1}{T_s} \sum_k \delta\left(f - \frac{k}{T_s}\right)$$

Q. Consider the following random signal:

$$X(t|s) = A(s) \cos[2\pi f_c t + \theta(s)]$$

where θ is a uniform R.V distributed in $[0, 2\pi]$ and A is also a uniform R.V distributed between $-a$ and $a > 0$. Is this process WSS?

$X(t|s) \rightarrow$ always a cosine with frequency f_c

Two things are random:

A and phase $\theta \rightarrow$ uniform b/w 0 and 2π
 $\downarrow$
uniform b/w $-a$ and a

so a cosine of some f_c but random height and random horizontal shift.

A process is WSS if mean is constant over time

$E[X(t)]$ does NOT depend on t

and Autocorrelation depends only on time difference τ .

$$R_x(t_1, t_2) = E[X(t_1)X(t_2)]$$

must reduce to

$$R_x(\tau) \quad \text{where } \tau = t_2 - t_1$$

Check WSS:

mean:

$$E[X(t)] = E[A \cos(2\pi f_c t + \theta)]$$

because A is independent of θ :

$$E[X(t)] = E[A] E[\cos(2\pi f_c t + \theta)]$$

Compute $E[A]$:

$$A \sim U[-a, a]$$

mean is 0:

$$E[A] = 0$$

So:

$$E[X(t)] = 0, E[\cos(\cdot)] = 0$$

mean is constant.

Autocorrelation:

$$X(t+T) = A \cos(2\pi f_c(t+T) + \theta)$$

$$X(t) = A \cos(2\pi f_c t + \theta)$$

$$X(t)X(t+T) = A^2 \cos(2\pi f_c t + \theta) \cos(2\pi f_c t + 2\pi f_c T + \theta)$$

$$\text{Using } \cos u \cos v = \frac{1}{2} [\cos(u-v) + \cos(u+v)]$$

$$X(t)X(t+T) = \frac{A^2}{2} \cos(2\pi f_c T) + \frac{A^2}{2} \cos(4\pi f_c t + 2\pi f_c T + 2\theta)$$

$$R_X(T) = E[X(t)X(t+T)]$$

$$R_X(T) = \frac{1}{2} E[A^2] \cos(2\pi f_c T) + \frac{1}{2} E[A^2 \cos(4\pi f_c t + 2\pi f_c T + 2\theta)]$$

$$R_X(T) = \frac{1}{2} E[A^2] \cos(2\pi f_c T)$$

$$E[A^2] = \frac{1}{2a} \int_{-a}^a A^2 dA$$

$$\downarrow$$
$$\left[\frac{A^3}{3} \right]_{-a}^a$$

$$= \frac{2a^3}{3} \Rightarrow \frac{1}{2a} \cdot \frac{2a^3}{3} = \frac{a^2}{3}$$

$$R_X(T) = \frac{1}{2} \cdot \frac{a^2}{3} \cos(2\pi f_c T) \Rightarrow \frac{a^2}{6} \cos(2\pi f_c T)$$

depends only on T .

It's a WSS process.